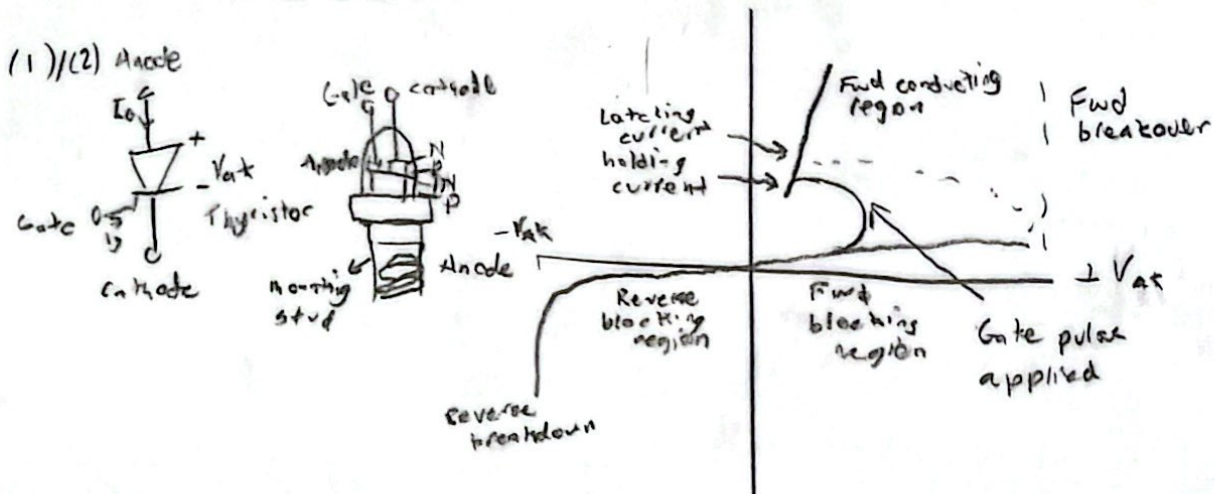




- (3)
- (a) The output DC voltage V_o can be controlled by changing the thyristor firing angle α , the delay angle with respect to input AC voltage V_s .
- (b) By maintaining a constant firing angle α

(4) Connect thyristor amplifier V_o to armature of the DC motor:

(i) $V_o = R I_o + E$ (ii) $T = k \phi I_o$

(iii) $E = k \phi \omega$

$E \rightarrow$ Motor Back-EMF, $I_o \rightarrow$ output/armature current, $R \rightarrow$ motor resistance
 $\omega \rightarrow$ Motor angular speed, $\phi \rightarrow$ motor flux, $k \rightarrow$ constant
 $T \rightarrow$ Motor torque

(a) V_o is a function of $\alpha \rightarrow V_o = f(\alpha)$. Controlling α controls V_o ; Controlling ω controls E .

(b) Feed back controls $\alpha \rightarrow$ to control V_o , ω controls E

(5)

(a) $\omega = \frac{V_o}{k\phi} - \frac{R}{k\phi} I_o$ (b) $T = k\phi I_o$ (c) $\omega = \frac{V_o}{k\phi} - \frac{R}{(k\phi)^2} T$

(6) Feed back controls $\alpha \rightarrow$ to control $V_o \rightarrow$ to control $E \rightarrow$ to control ω

(7) Restriction i.
 Motor speed control by controlling armature voltage is restricted by the rate of armature voltage increase as the increase of the motor EMF is proportional to the motor speed which is restricted by the motor inertia. Too fast increase of armature voltage may result in excess current flowing into the motor as the current is proportional to the difference between the armature voltage and EMF.

and

Restriction 2:

Motor speed control by motor field is restricted by not lowering the field too much; otherwise excess overspeed may result.

(8) Generated EMF in the armature of a DC machine is a function of ω_m and ϕ , and some physical parameters of the machine including number of P and conductors in the armature, and parallel paths in a .

$$E = \frac{PZ}{2\pi a} \phi \omega_m = K_e \omega_m = K_f \omega_m$$

$K_e \rightarrow$ voltage constant of the machine for a given air-gap flux.

Developed torque is a function of I_a , ϕ as well:

$$T_a = \frac{PZ}{2\pi a} \phi I_a = K_a \phi I_a = K_t I_a$$

$K_a \rightarrow$ torque constant of the machine for a given air-gap flux.